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Research paper

Hybrid machine learning and deep learning framework for multi-stage fault diagnosis in modern power systems

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ABSTRACT

Modern power system especially smart grids, integrating with different renewable energy resources, distributed generations, communication networks, and control systems, operate in islanded or grid-connected modes becoming more complex with time. This introduces a new challenge in faults detection during transition of modes which leads to changing behavior and characteristics of faults. This paper proposed a AI-based multi-stage fault diagnosis framework for modern power system by presenting hybrid random forest and convolutional neural network (HRF–CNN) Model. Unlike previous approaches relay on single model for faults detection, classification and localization (FDCL). Proposed hybrid model is not used only for detection of faults also determine Islanded or Grid-Connected mode of operation in smart grid. In RF model, which consists of two output layers, one for detection of faults to determine grid status (Normal or Fault), as well as a second one classify the operational mode of the smart grid (Islanded or Grid-Connected). If a fault occurs as recognized by the RF, then moves to two distinct CNN models separately. The first classifies a classification of faulty bus, while the second use for the classification of faults from among a set of 11 predefined short circuit faults. We employed Modified IEEE 14-Bus system integrated with renewable energy sources on MATLAB/Simulink to generate varied fault datasets as well as to carry out extensive simulations under different fault scenarios. The outcome validates that the integrated system accurately detects grid status, determines smart grid mode, detects the location of faults, as well as classifies types of faults in grid connected as well as in islanded modes. The proposed framework achieves 98% accuracy of fault detection, 95% accuracy of mode identification, 80% accuracy of fault bus localization, and 87% accuracy of fault classification, outperforming conventional machine learning and deep learning models such as SVM, KNN, and RNN. The results demonstrate that the HRF–CNN framework provides a scalable and efficient solution for fault diagnosis in complex smart grid environments.

1. Introduction

Modern power systems are getting more attention because of the expanding use of renewable energy sources, including wind energy, solar photovoltaics (PV), electric vehicles (EV) and battery energy storage systems (BESS) (Salehimehr et al., 2024). Global initiatives to create smart cities, which use integrated infrastructure, digital innovation, and environmental sustainability to enhance urban life, have been spurred by the rising urbanization of the world (Habib et al., 2025; Ehsan et al., 2025; Habib, 2025). A smart grid can function flexibly in both islanded and grid-connected modes, guaranteeing continuity of power supply under various operating conditions. However, this dual-mode operation also presents major challenges for system monitoring, fault diagnosis and protection due to the dynamic interactions among distributed generators, network topology, and control systems (Chandak and Rout, 2021; Salam et al., 2008).

Traditional fault diagnosis approaches often rely on model-based or single data-driven techniques. Consequently, there is a growing need for intelligent and adaptive diagnostic frameworks that can efficiently

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List of Acronyms

Acronym	Full form	Acronym	Full form
AI	Artificial Intelligence	ANN	Artificial Neural Network
AUC	Area Under Curve	BESS	Battery Energy Storage System
CNN	Convolutional Neural Network	CSV	Comma-Separated Values
DG	Distributed Generator	DL	Deep Learning
DT	Decision Tree	EV	Electric Vehicle
FDCL	Fault Detection, Classification, and Localization	FDC	Fault Detection and Classification
IIDG	Inverter-Interfaced Distributed Generation	KNN	K-Nearest Neighbors
LG	Line to Ground Fault	LL	Line to Line Fault
LLG	Double Line to Ground Fault	LLL	Three Phase Fault
LLLG	Three-Phase to Ground Fault	LSTM	Long Short-Term Memory
ML	Machine Learning	PCA	Principal Component Analysis
PMU	Phasor Measurement Unit	PV	Photovoltaic
ReLU	Rectified Linear Unit	RF	Random Forest
RNN	Recurrent Neural Network	ROC	Receiver Operating Characteristic
SG	Smart Grid	SVM	Support Vector Machine
GNN	Graph Neural Network	WT	Wavelet Transform

identify, classify, and localize faults under diverse operating modes of the system (Badihi et al., 2019). Reducing power transmission line renewal costs and improving the possibility of SGs protection are two benefits of accurately detecting and classifying transmission line faults. Transmission line failure causes consumers' electricity to be disrupted (Anwar et al., 2025).

Transmission lines (TL) are vital part of the power systems that carry electrical energy from generating facilities to distribution systems and, finally to end users, allowing for dependable and effective long-distance power transmission (Gönen et al., 2024). The faults happen irregularity at various location because it is not feasible or practical to design and construct an electric power system that would totally preclude the possibility of these faults occurring (Hewitson et al., 2004). Mostly power TL faults are caused by external factors, like exposure to the environment and contact with living things. Approximately 70% of all disturbance and blackouts in electric power systems are caused by short-circuit failures that significantly affect users (Chen et al., 2019).

Open circuit faults and short circuit faults are the two primary categories of faults in a power network. One or more conductor failures are the primary cause of open circuit faults in the system. There are two primary categories of short-circuit faults: symmetrical and asymmetrical (Ustun et al., 2021). These defects can be caused by a variety of factors, including bad weather, malfunctioning machinery, human mistake, etc. The faults in SGs can be erratic and require a targeted approach to repair the affected parts (Eslami et al., 2017). The SGs supply line can experience a variety of short circuit faults, which result in an unstable current signal (Prasad et al., 2018). short circuit faults cause instability in the power system and have an impact on power quality (Magdy et al., 2019; Al Hassan et al., 2018). When these faults appear, the system's functioning becomes abnormal. It may actively affect the operating voltage and pose a risk to both humans and animals. The equipment may overheat because of abnormal currents due to electric faults. The faults detection and classification (FDC) is a major problem. Also, the location of the fault must be identified as soon as possible in order to isolate faulty parts from the primary healthy system. Several fault analysis systems are examined to classify the faults in SG (Fahim et al., 2019, 2020).

In the literature, the firstly prominent methods such as artificial neural networks, fuzzy logic approaches, and wavelet transformation (WT) based analysis are presented. Then, to increase detection accuracy and resilience, hybrid techniques combine various combinations of these basic methods with data-driven techniques like PCA and PMU based methods. These days, cutting-edge techniques like SVM and advanced AI-based models have drawn a lot of attention lately (Mukherjee et al., 2021).

AI approaches are widely utilized for fault diagnosis along with how they cooperate in smart grids and power systems (Zahraoui et al., 2024). Modern power systems relies on ML and DL for information processing. The ML technique used a different learning algorithm to manage labeled data and apply fault diagnosis in TL. The amount of data in timeseries format is increasing as a result of sensor data collection and online activity. The related waveforms of voltage and current resulting from a short circuit are multivariate timeseries, or sequences of values. As a result, timeseries fault classification can be used to solve a number of relevant power system challenges (Nawaz et al., 2024).

Four ML classifiers with HP tuning are proposed in a paper (Bhattacharya and Nigam, 2023) for fault detection and classification of TL. The study (Ogar et al., 2022) suggested a categorical boosting classifier to classify electric faults effect the TL using the information of currents and voltages that are used to train the classifier. It recommended more optimization for autonomous training and real-time data mining. Using infrared thermal cameras, a non-invasive computer vision system is presented in Singh et al. (2021). An automatic FDC system for two TL configurations with ten different types of faults is presented in this paper (Goni et al., 2023) applying the novel ELM using fault resistance and fault signal per-unit values considered as features for the dataset compilation. Livani and Evrenosoğlu (2013) used discrete DWT to recover transient information from the voltage signals, then applied an SVM classifier, and wavelet coefficients helped in locating faults. The LSTM for FDC in TL by processing operational data directly without the need for intricate feature extraction (Rafique et al., 2021). The Intrusion detection capability is increased by adjusting the model's ideal weights (Kagade and Vijayaraj, 2024).

As aforementioned, there is a knowledge gap regarding SG fault detection classification and localization techniques. A single AI model is not sufficient for all purposes; each model is useful for specific task. As it does not exploit the strengths of different ML and DL methods fitted to specific faults diagnostic tasks. Moreover, most studies have focused only on three diagnostic parameters (FDC) without considering the operational mode detection of the smart grid. Therefore, there exists a research gap in developing a comprehensive and hybrid diagnostic framework that can simultaneously handle multiple diagnostic tasks including fault detection, classification, localization, and mode detection, under different operating conditions of the smart grid.

The contributions of this paper are listed as follows:

- This study proposed a hybrid intelligent diagnosis framework designed specifically for fault analysis of modern power systems which utilized both machine learning and deep learning techniques to enhance diagnostic adaptability and robustness against nonlinear and dynamic system behaviors.

Table 1
Comparative analysis of proposed framework to others methods.

Ref	Model	Application	Fault detection	Fault classification	Mode detection	Fault location	Hybrid
Najafzadeh et al. (2024)	ANFIS	Power grid lines	✓	✓	✗	✓	✗
Goni et al. (2023)	CNN	Transmission Lines	✓	✓	✗	✗	✗
Aziz et al. (2025)	SVM	HV power system	✓	✓	✗	✗	✗
Elavarasi (2023)	GNN	Smart grids	✓	✓	✗	✗	✗
Kumar et al. (2024)	Random Forest	Industrial Machinery	✓	✗	✗	✗	✗
Chen et al. (2024)	RNN	Power grid	✓	✗	✗	✓	✗
Mansouri et al. (2022)	RNN	Wind Energy	✓	✗	✗	✗	✗
Veerasamy et al. (2021)	LSTM	PV system	✓	✗	✗	✗	✗
Shadi et al. (2022)	LSTM	Distribution lines	✓	✓	✗	✓	✗
Belagoune et al. (2021)	LSTM	Transmission lines	✓	✓	✗	✗	✗
Alhanaf et al. (2024)	CNN–LSTM	Transmission lines	✓	✓	✗	✗	✓
Proposed Method	HRF–CNN	Transmission lines	✓	✓	✓	✓	✓

- The novelty of this work lies in the development of a HRF–CNN framework that performs multi-stage fault diagnosis, including fault detection, operational mode identification, faulted-bus localization, and fault-type classification within a unified sequential architecture rather than relying on a single AI model. Specifically, the RF model is employed for fast binary decision-making to determine system health status and operating mode, while two CNN models are used for higher-level multi-class tasks, namely faulted-bus localization and fault-type classification.
- The framework proposed novel diagnosis dimension, focused on mode identification which determine the smart grid is operating in island or grid connected mode as addressed in Table 1, which includes mode detection (grid-connected vs islanded) as a dedicated diagnostic task, which is rarely addressed jointly with FDCL in prior studies. Accurately determining the operational mode is critical because system behavior, fault signatures, and protective decisions vary significantly between the two modes.
- The proposed approach is tailored for complex and dynamic smart grids, addressing the increasing structural and operational complexity of modern smart grids, which include distributed generation, renewable sources, storage units, and grid interconnection, which makes the framework more practical for present-day power systems. It provides valuable insights for operators and contributing to safer, more stable, and intelligent grid operation.

The structure of the paper is as follows: Section 1 introduced the literature review, research gap, and paper contribution. Section 2 explained the modern power system integrated with renewable energy sources and fault types in the power system. Section 3 discusses the AI models, such as RF and CNN, used in this study. Section 4 presented the proposed methodology including the simulation setup of modified IEEE 14 bus power system, data collection and preparation, proposed AI-based framework and algorithm, and evaluation of model performance. Section 5 discussed the results of proposed model. Section 6 concluded the achievements, areas for improvement, and direction for future work.

2. Design of test modern power system

A modern power system particularly smart grid (SG) is made up of several interconnected renewable energy resources, distributed energy generations, communication networks, and control systems that work together to produce, store, and distribute electricity (Ahmad et al., 2025a). Its architecture offers energy resilience, sustainability, and effective power management by operating in both grid-connected and islanded modes (Saeed et al., 2021). A SG is basically a small scale control center of electricity production and use which can be de-coupled to the main grid and can work in its own capability.

The smart-grid system integrated with multiple generations and load types is selected because it better reflects the operational complexity of modern power systems. Contemporary smart grids are no longer based on conventional centralized generation only. Therefore, a selected modified IEEE 14-bus system include renewable energy resources including PV systems and wind turbines, BESS, distributed generators such as diesel units, and dynamic load categories. This integrated configuration is intentionally selected to ensure that the developed fault-diagnosis framework is evaluated under realistic and practically relevant operating conditions. The inclusion of multiple generation sources and load variations also enables the framework to learn fault signatures under both grid-connected and islanded modes, which is one of the main objectives of the study.

The proposed framework is developed for a modern smart-grid environment with renewable generation, storage, and distributed sources, and therefore it is more representative of systems with high penetration of inverter-interfaced distributed generation (IIDG). Nevertheless, under extremely high IIDG penetration, protection behavior may change due to reduced fault current magnitude and altered transient characteristics. Therefore, broader validation under different renewable penetration levels is an important direction for future work.

Fig. 1 demonstrates a grid-connected SG system including a BESS, diesel generator, wind energy and solar power. Renewable energy generating includes wind and solar PV. The use of a battery ensures a steady power source for the load.

The design of mathematical formulation of the SG components is covered in the sections that follow Roy et al. (2023).

2.1. Solar power system

Solar energy is one of the primary renewable resources integrated into modern power systems. Accurate modeling of photovoltaic (PV) arrays is essential for predicting power output with different environmental conditions and for evaluating the performance of the smart grid components. The PV array cells generate solar PV power, which is defined as:

$$P_{\text{pv}} = \eta A G\,[1 - \alpha\,(T - 25)] \tag{1}$$

where η, A, G, T, and α are efficiency of solar panel, panel area, solar irradiation, temperature, and coefficient of power degradation, respectively.

2.2. Wind power system

Wind energy contributes significantly to sustainable electricity generation in smart grids. Modeling the wind power system allows assessment of energy availability and integration with other distributed resources for reliable grid operation. Wind energy generated by a wind system is recognized by regulation as:

$$P_w = \frac{1}{2}\sigma A_b v^3 \tag{2}$$

where σ, A_b, v are air density, blade area, and speed of the wind, respectively.

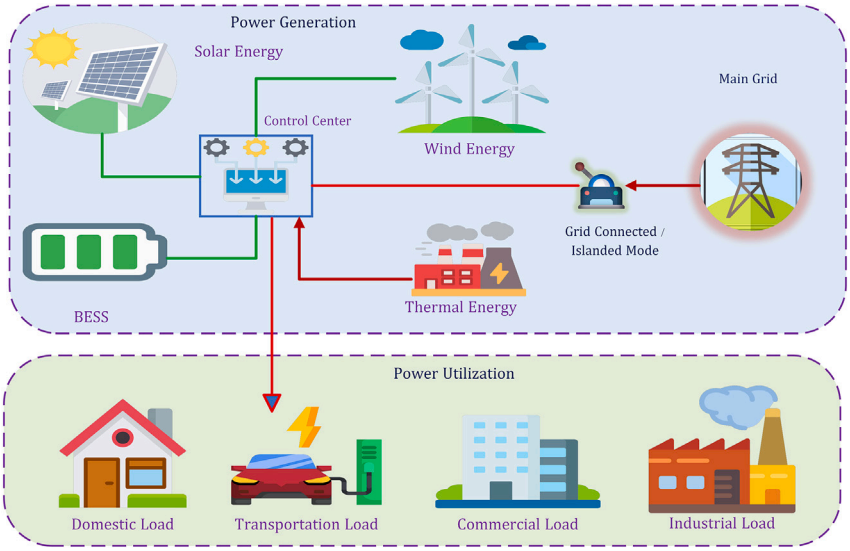


Fig. 1. Modern power system integrated with renewable energy sources.

2.3. BESS system

BESS provide flexibility and stability to the smart grid by balancing supply–demand mismatches. Proper modeling of battery voltage and state-of-charge is crucial for optimal operation and integration with renewable sources.

2.4. Distributed generation system

Diesel and other distributed generators ensure continuous power supply in hybrid smart grids, particularly during fluctuations of renewable energy generation. Modeling the output of these generators is necessary for load sharing and system stability analysis.

2.5. Faults in transmission line of power system

Electrical grid failures are grouped according to where they come from. The most frequent kind of problems in electricity systems are caused by natural causes, which mostly include disruptions caused by extreme weather events such as tree fall and broken insulators (Mar et al., 2019; Hare et al., 2016).

Microgrids experience different types of faults similar to traditional power systems. Understanding these faults in the context of microgrids, especially during switching transients, is crucial for ensuring reliability and stability. Inter-phase insulation breakdown may result from transient overvoltage caused by switching operations, such as connecting or disconnecting microgrid segments. A symmetrical short circuit may result from high switching transients if insulation fails in every phase (Mukherjee et al., 2021). Asymmetrical short circuit faults have smaller scale and possibly weaker security measures, microgrids are especially vulnerable to threats. Because switching transients produce uneven voltage and current spikes across phases, they can readily cause unsymmetrical failures (JUNAIDI et al., 2021).

Series faults generally involve conductor interruption or open-circuit conditions and are usually associated with current reduction, phase discontinuity, or unbalanced operating behavior. In contrast, shunt faults involve short-circuit paths between phases or between phase and ground, resulting in sudden current surges, waveform distortion, and severe transient behavior. The present work is focused on short-circuit shunt faults, specifically the eleven classes: AB, BC, AC, AG, BG, CG, ABG, BCG, ACG, ABC, and ABCG. The different shunt faults are detected and classified based on their distinct three-phase current signatures. All 11 categories of short-circuit faults are categorized as symmetric faults (ABC and ABCG) and asymmetric faults (AG, BG, CG, ABG, BCG, ACG, AB, BC, and AC), as indicated in Fig. 2. It should be emphasized that G stands for the ground, and A, B, and C for the three phases.

3. Proposed AI based models

The SGs are becoming attractive due to the fact that they enhance energy reliability and efficiency. But a major challenge is to the protection and safety of these systems (Ahmad et al., 2025b). Lately, AI has been proposed as an effective way of dealing with these issues. Protective schemes based on AI which is advanced algorithms that create the best fault diagnosis system possible in SGs regarding reliability and responsiveness. These algorithms can instantly identify anomalies or faults by analyzing data from multiple sources, including sensors and smart meters. Proposed AI models, RF and CNN Models, are used to detect faults in distributed energy resources and TL, allowing for prompt action to safeguard the SGs.

3.1. Random Forest (RF)

A RF is a supervised learning method which uses several trees for classification. RF, in the other word, combines the different decision tree (DT) to get more reliable and robust final prediction. RF are extremely effective machine learning algorithms that can easily fit extremely complicated data sets and do classification. Because RF method require relatively minimum data representation, the computational cost for fault classification can be greatly decreased. In RF, each classifier identifies an input vector by using a random vector that is randomly sampled. Each tree generates a unit vote for the corresponding class. The RF uses Gini Index, which measures the degree of impurity in an attribute matching to a class, as an attribute selection metric (Banire et al., 2024), which is expressed as follows:

$$\text{Gini Index} = 1 - \sum_{x \in X} P(x)^2 \tag{3}$$

The probability that class x belongs to a collection of classes X in the dataset is denoted by $P(x)$.

3.2. Convolutional Neural Network (CNN)

A CNN, a deep learning model extracts hierarchical characteristics from structured grid data such as images (2D), time-series signals (1D),

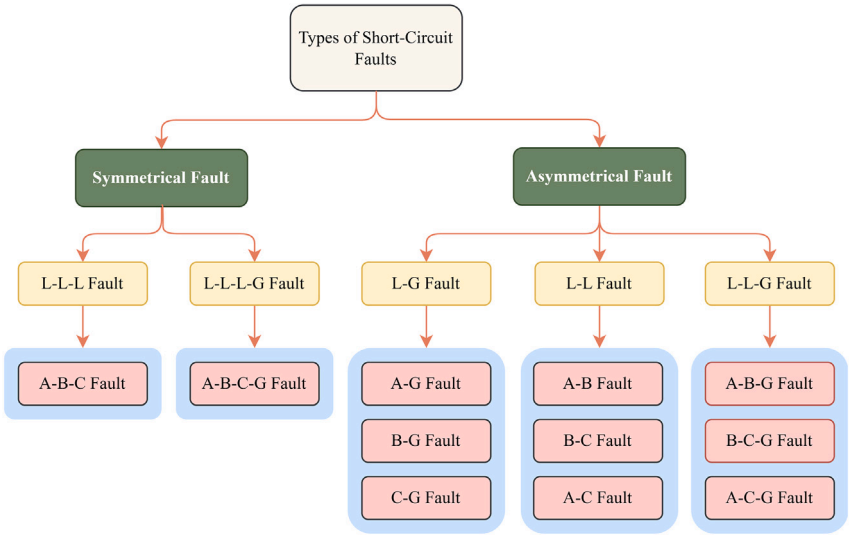


Fig. 2. Classification of power transmission line faults.

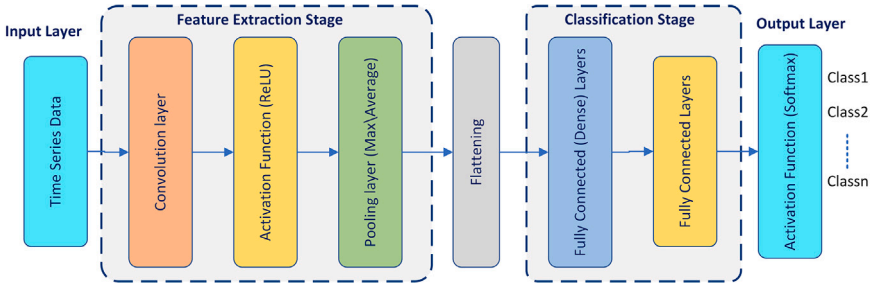


Fig. 3. CNN structure.

and spatio-temporal data (3D). It exploits local connectivity, weight sharing, and translation invariance to efficiently learn meaningful patterns from raw data. These characteristics make CNNs robust and highly effective for pattern recognition, and fault detection. In certain state-of-the-art research, the CNN is introduced to address the issues of uncertainty in the power TL (Guo et al., 2019; Pan et al., 2022; Wang and Chen, 2019). By effectively extracting the features from the time series data, the CNN based fault classification model demonstrates remarkable performance in fault classification.

CNN's main function is to extract spatial properties from data by using a variety of filters. The CNN's convolution layers are trained to extract highly discriminative features, and the latter layers mimic a multilayer perceptron that performs the classification task and CNN structure is shown in Fig. 3.

4. Methodology

4.1. Proposed hybrid AI model

This part talks about the basic design of the AI based FDCL frameworks, we created for SGs. It covers the main parts of the hybrid framework, what it aims to do, and how it all works together using deep learning and machine learning models. We really emphasize on the way this arrangement can identify, categorize and locate the fault in various circumstances. The purpose of the proposed research is to come up with a powerful, hybrid AI-based scheme to SGs to achieve rapid and precise fault detection, classification and localization during islanded and grid-connected conditions. This architecture combines machine learning (e.g RF) and deep learning (e.g CNN) into a sequential decision-making system, which can react to real-time operating conditions.

The proposed HRF–CNN framework illustrated in Fig. 4 presents a comprehensive fault diagnosis and protection architecture for modern smart grids integrating multiple distributed energy resources. Modern power systems consist of various generation units, including PV, wind energy systems, BESS, distributed generators, and the main grid, which are interconnected through a central control center. This control center continuously monitors system behavior by acquiring real-time electrical measurements. These measurements represent the fundamental data source for intelligent decision-making.

The proposed hybrid AI-based fault diagnosis framework that acquired time-series current data are first processed by the RF model, which performs two critical tasks simultaneously: (i) fault detection by distinguishing between normal and abnormal system states, and (ii) operational mode identification by determining whether the smart grid is operating in islanded or grid-connected mode. After identifying a fault, the framework executes two parallel CNN models. The former CNN classifies the types of fault, particular short-circuit fault out of 11 fault types that exist, whereas the latter CNN localizes the fault i.e. the faulty bus in the system. The results of these models are subsequently sent to the decision and control protection system, where faulty areas can be isolated and stable and resilient operation of the smart grid can be facilitated. This sequential and modular design guarantees high diagnostic accuracy, low computational cost and flexibility to complex and dynamic grid conditions.

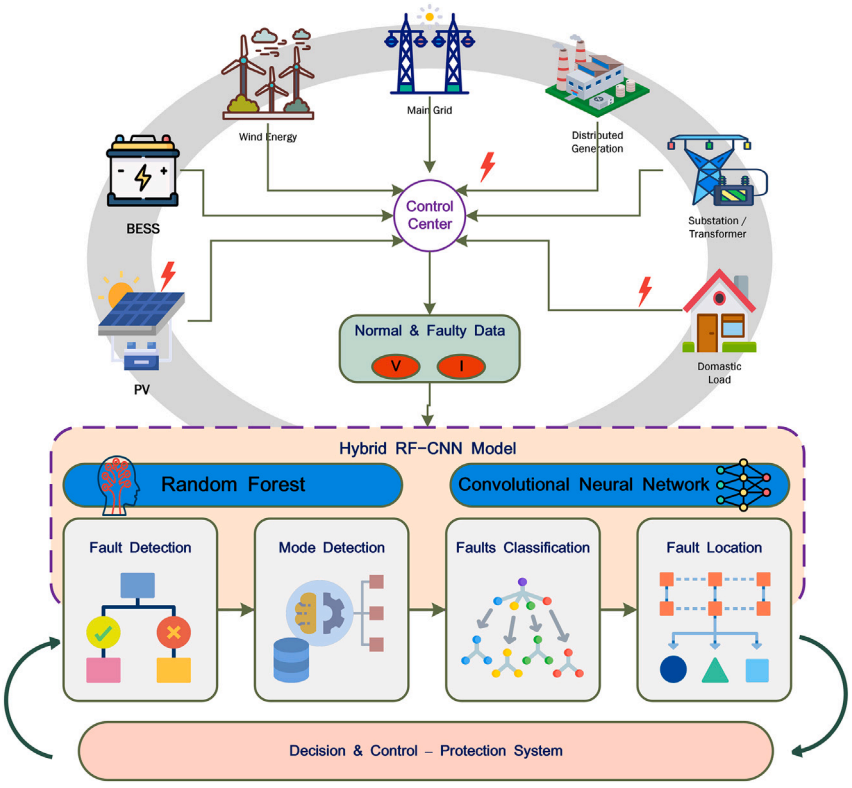


Fig. 4. Proposed HRF–CNN framework.

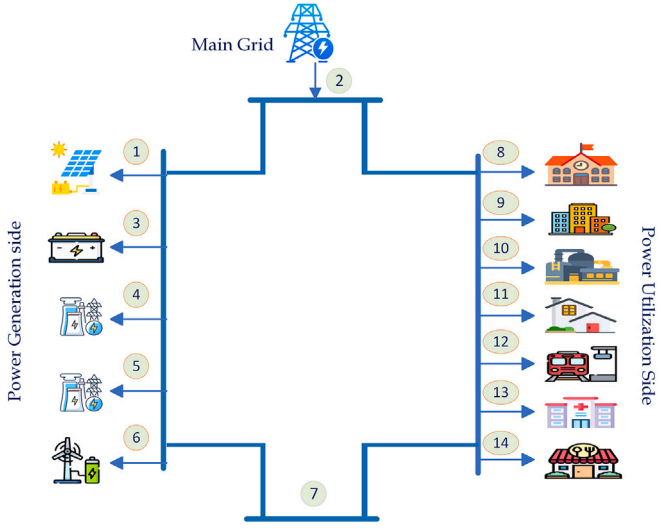


Fig. 5. Smart Grid System integrated with various generations and loads.

4.2. Simulation setup of test power network

A realistic and standardized smart grid test system is developed to validate the effectiveness and robustness of the proposed HRF–CNN fault diagnosis framework. The simulation setup is designed to emulate practical operating conditions of modern power systems, with high penetration of renewable energy sources, distributed generations, battery systems, and dual operating modes. A modified IEEE 14-bus system is selected as the benchmark network due to its widespread acceptance in power system research and its suitability for comparative performance evaluation. The test system enables systematic generation of diverse fault scenarios under both islanded and grid-connected modes, thereby providing a reliable platform for assessing fault detection, classification, localization, and operational mode identification capabilities of the proposed AI-based framework.

The whole smart grid structure, including all the generation units, load buses, and the transmission bus, have been simulated using MATLAB Simulink. IEEE 14-bus based structure guarantees standardized modeling and comparative analysis. The suggested modified IEEE 14-bus based SG system integrates various generation technologies such as solar, wind, diesel, battery storage, and grid interconnection, used as a test system as demonstrated in Fig. 5. This section provides a comprehensive descriptive and visual summary of the three dominant sections of the SG: the source side, the load side, and the TL.

Our power system incorporates six generation buses with diverse energy sources, summarized as follows:

- **Bus1:** Photovoltaic (PV) System — 3 MW
- **Bus2:** Grid Connection — 12 MW
- **Bus3:** Battery Energy Storage System (BESS) — 1 MW
- **Bus4:** Diesel Generator 1 (DG1) — 1.5 MW
- **Bus5:** Diesel Generator 2 (DG2) — 1.5 MW
- **Bus6:** Wind Turbine — 1 MW

These sources combine a hybrid energy mix that integrates renewable power with traditional power generation and storage capacity. This hybrid combination reinforces the resilience and sustainability of the SG.

The most important interface between the load and source sides is the TL of power systems. The end users or consumers of the generated electrical energy are the load side of the SG. There are seven buses that will serve various categories of load such residential, commercial and industrial consumers.

4.3. Dataset description

When trying to establish a comprehensive and sound fault detection and classification model of our microgrid, we have developed a large

dataset that would represent a wide range of operating conditions. We have added faulted and fault-free conditions to enable supervised learning and evaluation of classification algorithms. The initial consideration is made on voltage, current and frequency data but three-phase current is taken as the principal input feature because of its dynamic nature as a feature describing transient behavior. The sampling rate is decided to be a high temporal resolution in detecting transient events.

The data are organized under the top-level directory, which contains two primary status categories: Fault and Normal. Each of these categories is further divided into two operating modes, namely Grid-connected and Islanded. For the faulty condition, each mode includes six bus categories: BESS, DG1, DG2, GRID, PV, and WIND. Within each bus folder, the data are further classified into eleven fault types: AB, BC, AC, AG, BG, CG, ABG, BCG, ACG, ABC, and ABCG. The input features used in this study are extracted from the first three columns of each CSV file, representing the three-phase current signals, which are used as the primary inputs for the proposed machine learning and deep learning models.

Captured data is structured and saved in CSV format with appropriate labels for fault type, location, and mode. We considered a total of 11 distinct types of faults in our dataset. These include both symmetrical and unsymmetrical faults commonly observed in power systems. The dataset includes of CSV files, and each file holds time-series data gathered from the SG simulation model created in MATLAB/Simulink.

- **Fault-Included Data:** CSV files are created by simulating the SG under the above-described 11 types of faults. Each file is for a different instance or set of fault scenarios, such as varied locations, times, and levels of severity.
- **Normal Operation Data:** CSV files are created in normal operating conditions, free from the presence of any fault. These files are used as baseline data to enable the model to identify healthy and faulty states.

4.4. AI-based proposed framework

The SGs are becoming attractive due to the fact that they enhance energy reliability and efficiency. But a major challenge is to the protection and safety of these systems. Lately, Artificial Intelligence (AI) has been proposed as an effective way of dealing with these protection issues. Protective schemes based on AI are advanced algorithms that create the best protection systems possible in SGs with regard to reliability and responsiveness.

4.4.1. Model 1: Random Forest (RF) model for fault and mode detection

The RF is the first part of our hybrid protection system as shown in Fig. 4. It is responsible for detecting faults and figuring out what mode the SG is in. In this section, we described how the model is structured, how we trained it, and how we evaluated its performance using RF model to learn from the time-dependent features of the current signals.

4.4.2. Model 2: Convolutional Neural Network (CNN) model for identification of faulty bus

Once a fault is detected, the next task is to find out exactly where it is in the SG. This section introduces our second model, a CNN model, that helps pinpoint the faulty bus based on current waveform features. We shared details on how we trained the model, tuned its parameters, and the results we got, which showed how effective it is.

4.4.3. Model 3: Convolutional Neural Network (CNN) model for fault type classification

It is just as important to identify the fault type correctly to take the right protective actions. Here, we explore our third AI model, which uses the same data to classify the fault into one of eleven categories as illustrated in Fig. 4. We will detail the structure, training methods, and performance metrics of this model.

4.5. Integration of models and system flow

This section looks at how the three AI models come together in a smooth decision-making process. We will explain how everything operates in sequence and how it handles conditions in real-time. A flowchart is included to give a visual of the system architecture. In the first step of model execution sequence, the time-series current data is fed into the RF model for fault detection. If a fault is detected, the same input is then send to two separate CNN models in which one for fault location and the other for fault type classification as presented in Fig. 6. See the implementation of proposed HRF–CNN Model for FDCL in the Algorithm 1.

Algorithm 1 : Hybrid RF-CNN Model for Fault Detection, Classification and Localization

Require: System data: Normal and Faulty Data [Voltage $V(t)$, Current $I(t)$]

Ensure: Fault Detection, Mode of Operation, Classification of Fault, Location of Fault (Bus)

```
Collect real-time system data: [V(t), I(t)]
Give data into the RF Model
procedure RF PREDICT
    Operating mode ∈ {Grid-Connected, Islanded}
    Fault status ∈ {Fault, No Fault}
end procedure
if Fault status = No Fault then
    return Operating mode
else
    Feed same data to (CNN) Model
    CNN determines fault location (bus number)
    CNN classifies fault into one of 11 types
end if
return Operating mode, Fault location, Types of Faults
```

4.6. Evaluation of model performance

Performance of model is determined by using classification metrics, confusion matrices, and receiver operating characteristic (ROC) curve. This two-row, two-column matrix illustrates binary classification issues (Suliman et al., 2024). Fig. 7 demonstrates the confusion matrix's significance.

Accuracy, precision, recall, and F1-score are used to evaluate both classification tasks. They can be explained as follows Najafzadeh et al. (2024):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{6}$$

$$F1 - score = \frac{2 \times Precision \times Sensitivity}{Precision + Sensitivity} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{7}$$

The ROC curve is a graphical representation used to determine the performance of a classification model. It is plotted with sensitivity and the 1-specificity. Area Under Curve (AUC) of the ROC is scalar metric that measures the overall ability of a model to distinguish between classes. The larger AUC shows the better performance of model. Lee and Lin (2021)

4.7. Implementation of proposed methodology

To put our proposed framework into action, we used a mix of simulation tools and machine learning libraries are listed in Table 2.

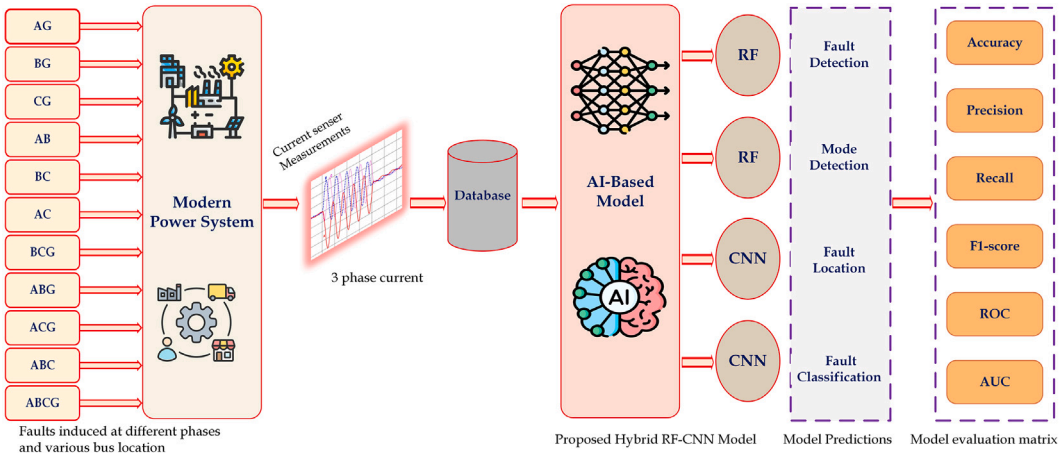


Fig. 6. Flow diagram of proposed model.

Table 2
Tools and technologies used.

Software	Libraries/Frameworks	Hardware
MATLAB/Simulink for Simulation Python for AI Model Training	TensorFlow/Keras for CNN Scikit-learn for RF	Intel(R) 10 Core(s) 20 Logical Processor(s) 64GB RAM

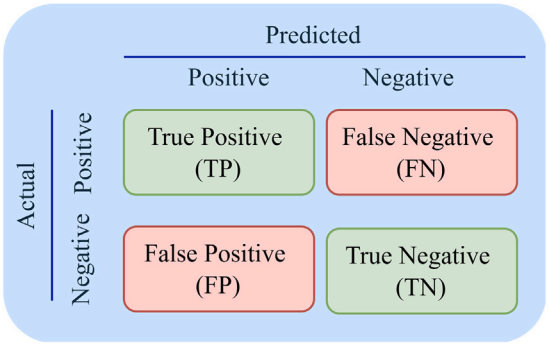


Fig. 7. Confusion matrix.

Table 3
CNN architecture for faulted-bus and fault-type classification.

Layer	Configuration
Input Layer	Shape: (500, 3)
Conv1D Layer 1	64 filters, Kernel size = 3, Activation = ReLU
Conv1D Layer 2	32 filters, Kernel size = 3, Activation = ReLU
Flatten Layer	Converts feature maps to 1D vector
Dense Layer	64 neurons, Activation = ReLU
Dropout Layer	Dropout rate = 0.2
Output Layer	Softmax activation (number of neurons = number of classes)

Table 4
CNN training parameters.

Parameter	Value
Loss Function	Sparse Categorical Cross-Entropy
Optimizer	Adam
Number of Epochs	20
Batch Size	64

Table 5
RF model configuration.

Parameter	Value
Number of Trees (n_estimators)	200
Maximum Depth (max_depth)	15 (for bus/fault classification)
Random State	42

5. Results and discussion

The modified IEEE 14 bus single line system as shown in Fig. 8 is simulated to generate the dataset. All 11 types of faults are introduced at all source buses and values of current and voltage are recorded for evaluation of type and location of fault. Current and voltage wave shape of source buses under normal and three phase fault condition are given below.

The output currents of source buses under normal condition in our modified IEEE 14-bus-based power system are given below in Fig. 9:

The output currents of source buses during fault at bus1, bus2, bus3, bus4, bus4, bus5, bus6 in our modified IEEE 14-bus-based power system are given below in Figs. 10, 11, 12, 13, 14, 15:

5.1. Training and validation

This dataset is used for training, validation, and testing of fault classification models. In this study value of currents are utilized for the detection, classification and identification of faulty bus. The richness of fault type diversity, along with normal data inclusion, ensures that the model gets trained to detect faults precisely without producing false positives when operating in normal conditions.

The input signals consist of the first three columns from each CSV file, corresponding to the three-phase current signals. For the CNN-based bus and fault-type classification tasks, sliding-window segmentation is performed with a window size of 500 samples and a step size of 100 samples. However, each segmented sample is structured into a matrix of size (500, 3), where 500 represents the number of time steps and 3 corresponds to the three-phase currents. The CNN model then directly processes these structured current sequences. For RF-based fault and mode detection, the data are segmented using a sequence length of 500 samples and a stride of 250 samples. The dataset is divided using an 80% training and 20% testing split, and for CNN training, 20% of the training set is used as validation data.

The CNN architecture and corresponding training parameters used for faulted-bus and fault-type classification are summarized in Table 3 and Table 4, respectively, while the configuration of the RF model is presented in Table 5.

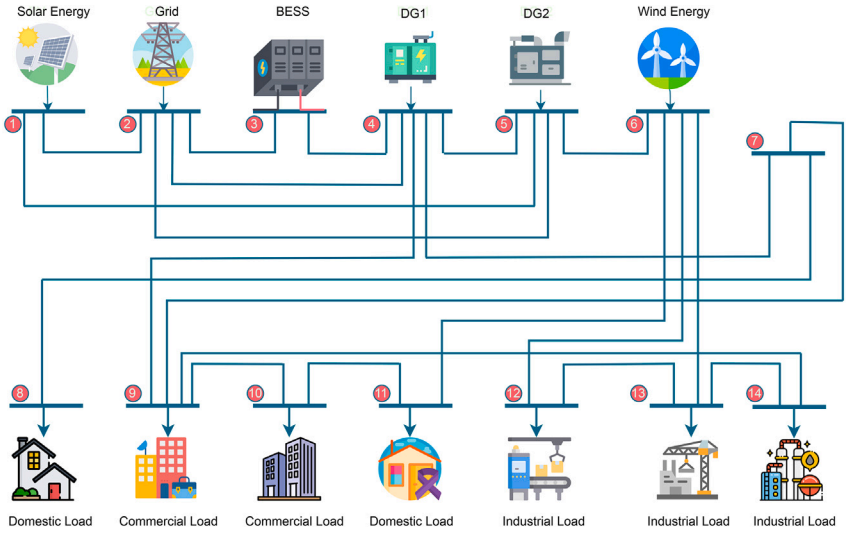


Fig. 8. Modified IEEE14 bus single line system.

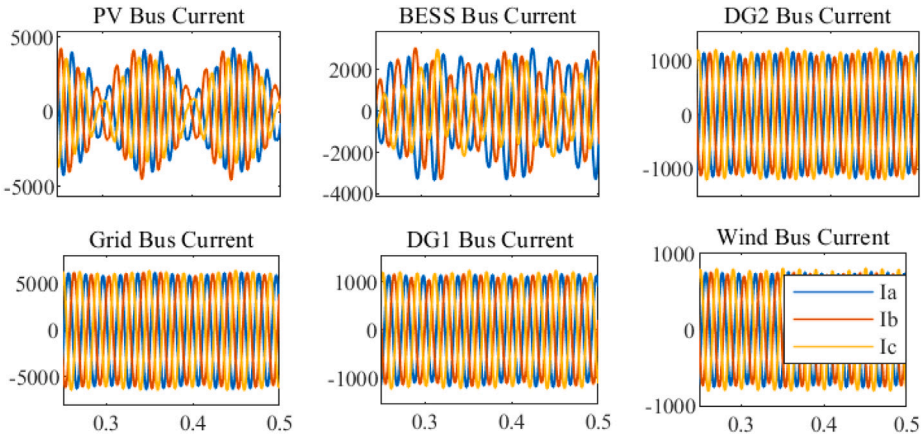


Fig. 9. Currents of generation buses during normal condition.

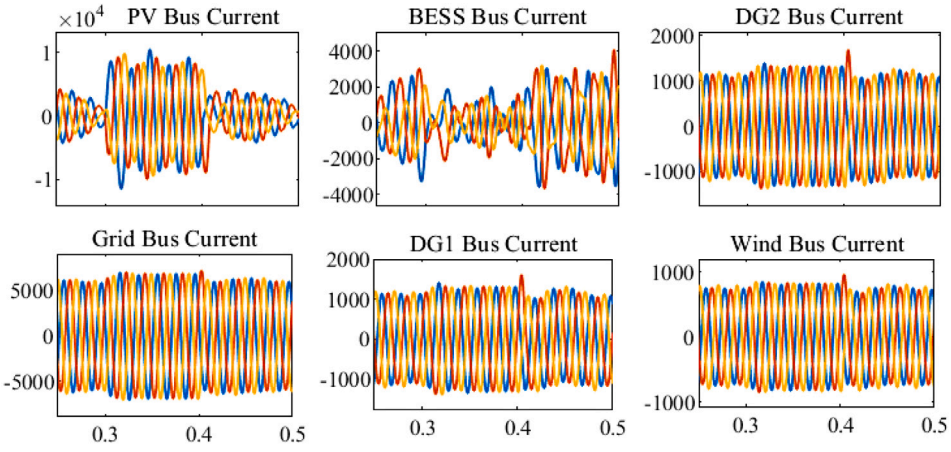


Fig. 10. Currents of generation buses during fault at Bus1.

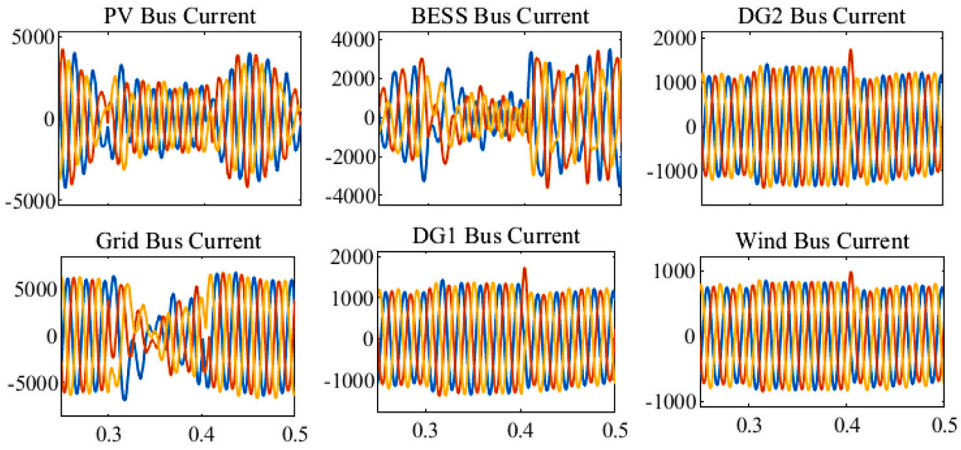


Fig. 11. Currents of generation buses during fault at Bus2.

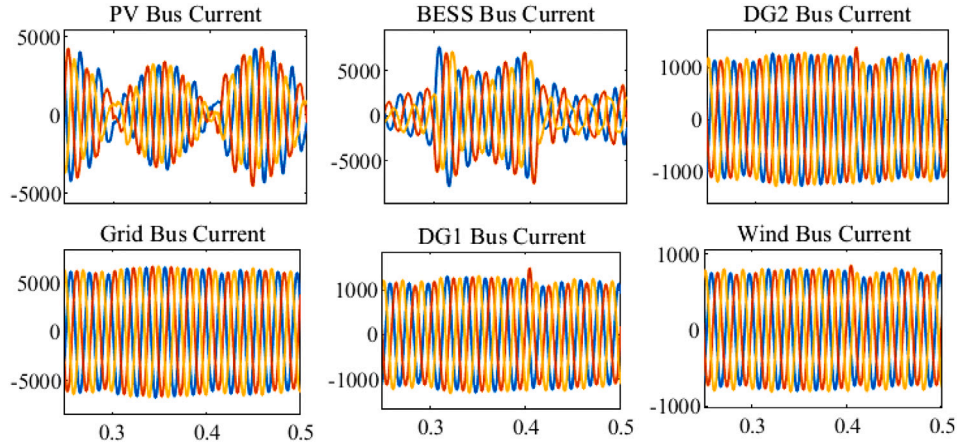


Fig. 12. Currents of generation buses during fault at Bus3.

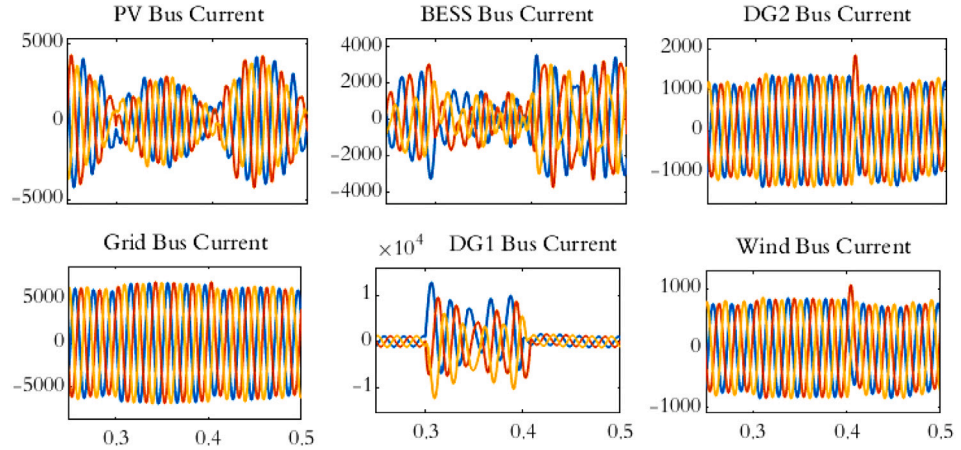


Fig. 13. Currents of generation buses during fault at Bus4.

Training is performed using time-series data from CSV files labeled with binary classes: faulty and normal condition. Performance of model in terms of accuracy and loss, is represented in Fig. 16.

5.2. Model 1: Normal and fault detection model using RF

In order to categorize whether a present data is a representative of a fault state or not, we used a RF classification model. RF models are naturally geared towards sequential data such as time-series signals since they have the capability of maintaining memory of past inputs, which assists in capturing the temporal dynamics inherent in electrical waveforms during fault occurrences. Fault detection performance using RF model and comparison with other Ml and DL models shown in Fig. 17. The performance of model, as illustrated by the confusion matrix and the ROC curve demonstrated its exceptional capabilities in Fig. 17.

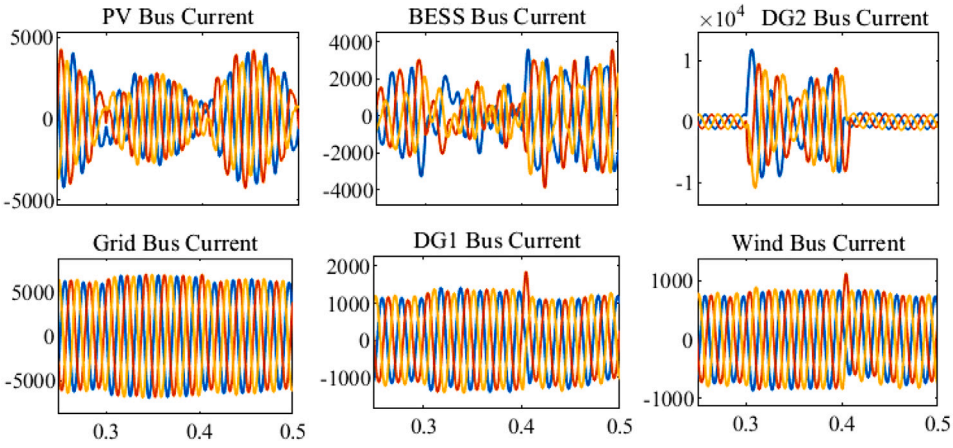


Fig. 14. Currents of generation buses during fault at Bus5.

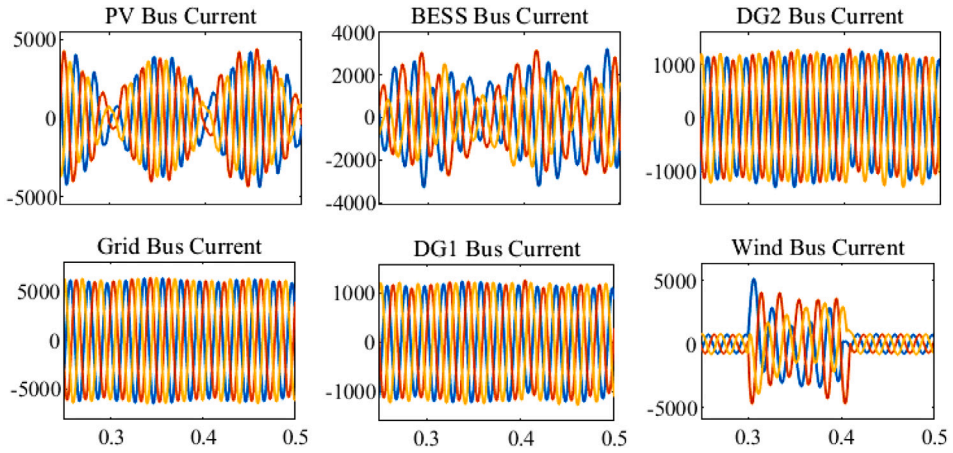


Fig. 15. Currents of generation buses during fault at Bus6.

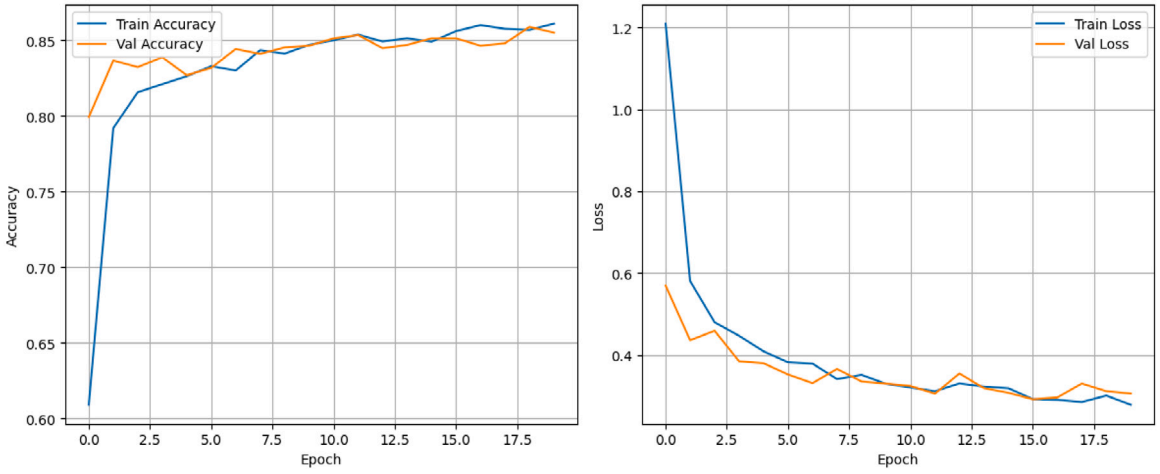


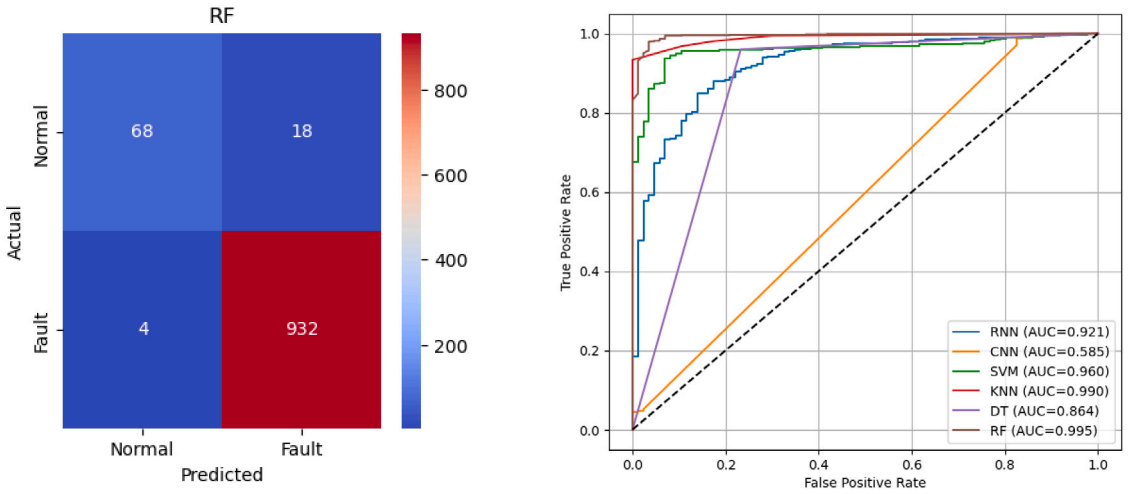
Fig. 16. Training and validation accuracy & loss.

5.3. Model 1: Detection of operating mode: Grid-connected vs islanded

Besides fault detection, the RF model is also trained to identify the smart grid's operational state as grid-connected or islanded mode. The mode detection performance using CNN model and comparison with RF Model shown Fig. 18. The model's effectiveness, shown through the confusion matrix and the ROC curve in Fig. 18, highlights its strong performance.

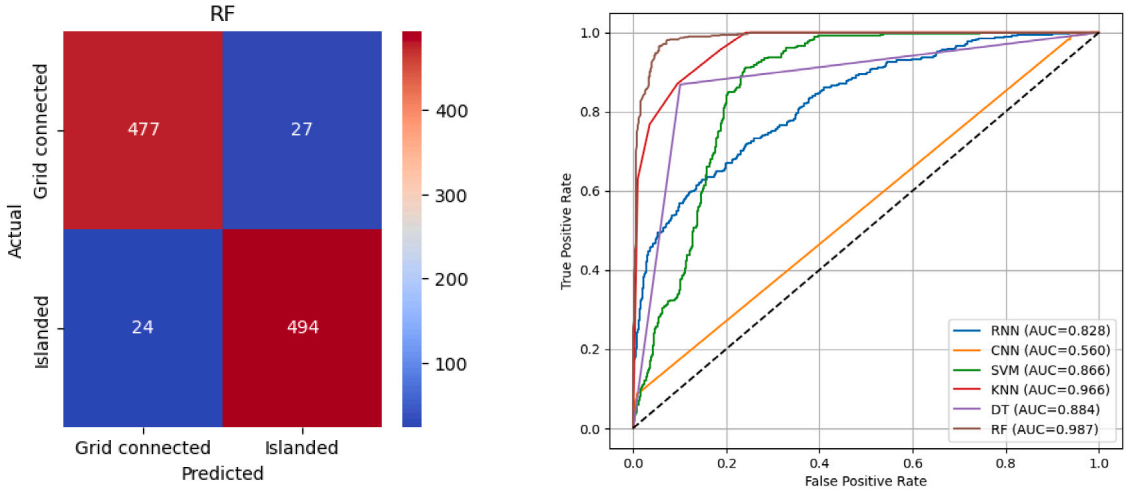
5.4. Model 2: Classification of faulty bus using CNN

The second model constructed in our research utilizes the CNN model algorithm, created to identify the position of a fault by determining the particular bus in the SG at which the fault has emerged. This model is instrumental in improving system protection and restoration, as rapid faulty bus is imperative for effectively isolating and resolving system disturbances. Model 2 deals with multi-class classification, with



(a) (b)

Fig. 17. Fault Detection (a) Confusion matrix of RF model, (b) ROC curve comparison of RF and other AI models with corresponding AUC values.



(a) (b)

Fig. 18. Mode Detection (a) Confusion matrix of CNN model, (b) ROC curve comparison of CNN and RF models with corresponding AUC values.

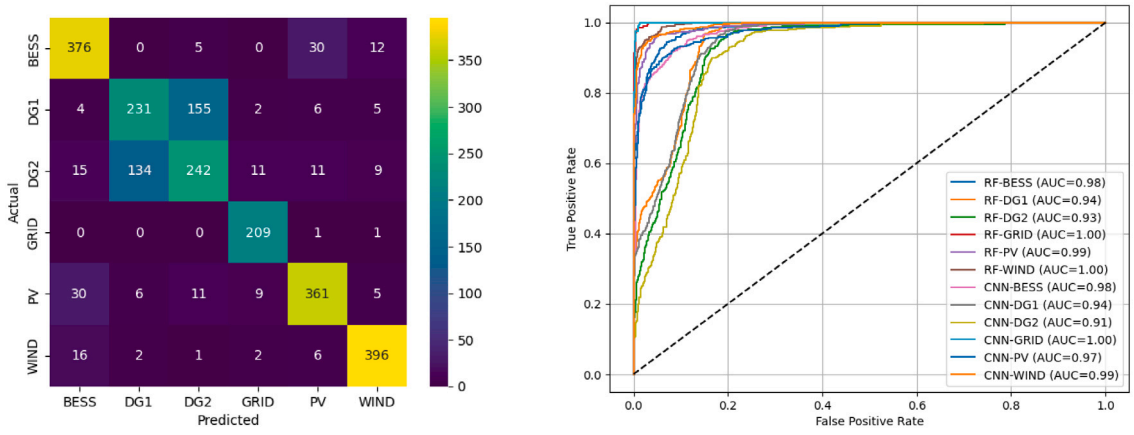
every class indicating a bus that belongs to a generation unit within the SG. Classification of faulty bus performance using CNN model and comparison with RF Model shown in Fig. 19. The confusion matrix, supported by the ROC curve and associated AUC, demonstrates the performance of the CNN model as shown in Fig. 19.

5.5. Model 3: Fault type classification using CNN

Model 3 uses the CNN algorithm to predict the fault type that arises in the smart grid. Identification of the fault type is important for protection systems because different faults impose different effects on system stability, coordination of protection, and restoration process. This classifier is trained to identify and classify 11 various electrical fault types according to signal features. Fault types classification performance using CNN model and comparison with RF Model shown in Fig. 20. The model's performance is evidenced by the confusion matrix as well as the ROC curve and its AUC value, is demonstrated in Fig. 20.

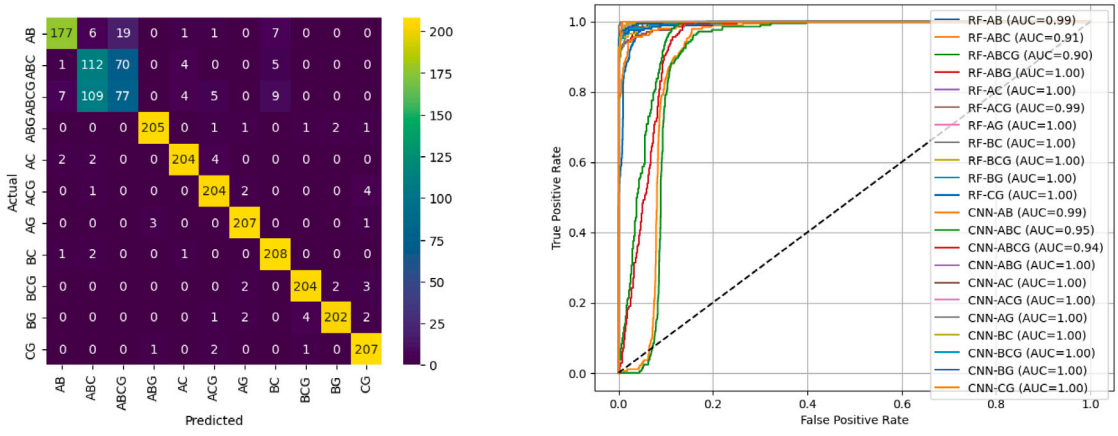
The performance analysis by the classes indicates that some types of faults, especially ABC and ABCG, have relatively lower precision and recall than other types of faults. This tendency because the waveform patterns become more similar as the high-order symmetrical fault conditions, which decreases inter-class separability. Moreover, the transient signatures of these fault classes are overlapped with other severe faults, and it is more difficult to distinguish between them. This can also be seen with some classes of bus like the DG1 and DG2 where an electrical response can be quite similar as they have similar operating conditions and position in the network. These results indicate that, with further signal channels (e.g. voltage, frequency, derived time–frequency features) added, further enhancements in future performance can be made.

The accuracy of the proposed multi-model framework is compared to other AI models and shown in Fig. 21, which demonstrating the high performance of the proposed framework. The comparative performance indicates that RF model performed better than SVM, DT, and KNN in fault detecting and mode detecting. Through the ensemble nature of RF, which is a combination of several decision trees to enhance generalization and minimize overfitting. RF also supports high-dimensional flattened time-series inputs, and can successfully learn nonlinear relationships without a large amount of parameter hunting. By contrast, SVM and KNN are more sensitive to the structure of feature space and might not be able to handle large, high-dimensional signal windows. In the multi-class tasks, the CNN model is found to be more successful in fault-type classification and faulted-bus classification since it naturally



(a) (b)

Fig. 19. Classification of Faulty bus (a) Confusion matrix of CNN model, (b) ROC curve comparison of CNN and RF models with corresponding AUC values.



(a) (b)

Fig. 20. Fault types classification (a) Confusion matrix of CNN model, (b) ROC curve comparison of CNN and RF models with corresponding AUC values.

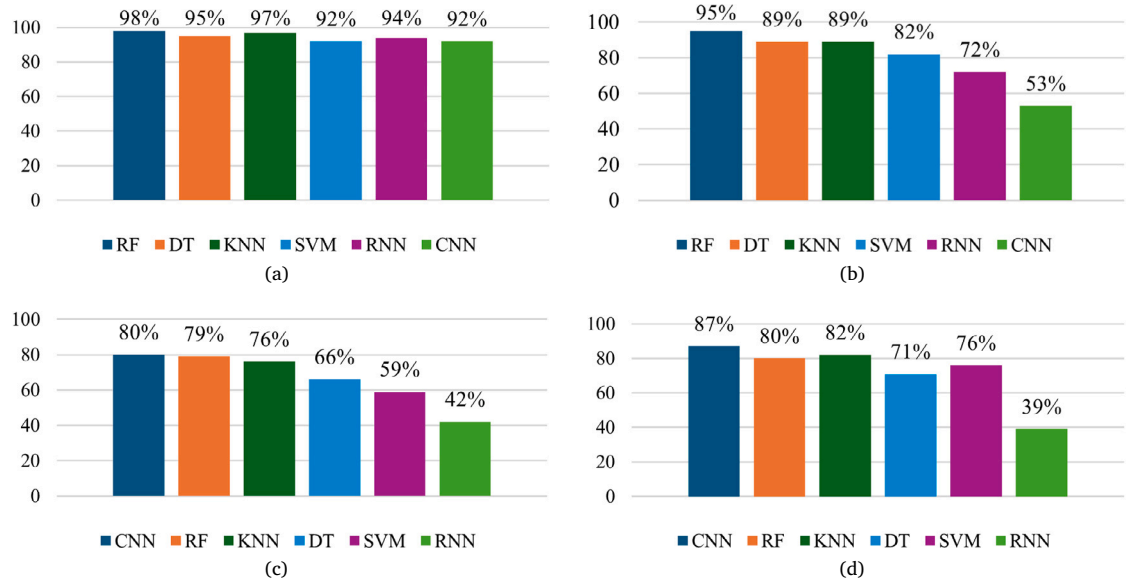


Fig. 21. Comparative accuracy of the proposed framework with different AI models: (a) Fault detection, (b) Mode detection, (c) Faulted-bus classification, and (d) Fault-type classification.

Table 6
Classification report of proposed framework.

Model:		**RF**		
		Precision	Recall	f1-score
Fault Detection status	Normal	0.94	0.79	0.86
	Fault	0.98	1.00	0.99
	Accuracy			**98%**
Mode of Operation	Grid Connected	0.95	0.95	0.95
	Islanded	0.95	0.95	0.95
	Accuracy			**95%**
Model:		**CNN**		
		Precision	Recall	f1-score
Classification of Faulty Bus	BESS	0.85	0.89	0.87
	DG1	0.62	0.57	0.60
	DG2	0.58	0.57	0.58
	GRID	0.90	0.99	0.94
	PV	0.87	0.86	0.86
	WIND	0.93	0.94	0.93
	Accuracy			**80%**
Classification of Faults	AB (LL)	0.94	0.84	0.89
	ABC (LLL)	0.48	0.58	0.53
	ABCG (LLLG)	0.46	0.36	0.41
	ABG (LLL)	0.98	0.97	0.98
	AC (LL)	0.95	0.96	0.96
	ACG (LLG)	0.94	0.97	0.95
	AG (LG)	0.97	0.98	0.97
	BC (LL)	0.91	0.98	0.94
	BCG (LLG)	0.97	0.97	0.97
	BG (LG)	0.98	0.96	0.97
	CG (LG)	0.95	0.98	0.97
	Accuracy			**87%**

learns local temporal patterns using segmented three-phase current signals. CNN also retains signal structure and extracts hierarchical features unlike traditional ML models which use flattened representations and are more informative to separate complex fault signatures.

Moreover, Table 6 has provided a comprehensive report of the performance metrics of the models, such as accuracy, precision, recall, and F1 score. The RF model had an accuracy of 98% on the test data. This implied that this model is accurate in classifying a condition to be faulty or normal. The ability to learn based on time sequences renders RF model to be better than the other traditional classifiers in dynamic signal classification. The RF model is found to have a 95% detection in the grid-connected or islanded operating mode of the SG. The CNN model accuracy level of bus location classification is 80% that confirmed its effectiveness and viability in intelligent protection systems. The CNN model used to classify fault types had train accuracy of 87%, which implied that it possesses strong potentials to extrapolate on a wide range of fault cases with various properties.

The framework suggested is scalable due to its modular nature. The initial step fast binary detection by RF, and further CNN models are only activated for classification, once a fault has been identified. This lowers redundant computation when carrying out detection operation and facilitates effective implementation in a bigger system. Even though the current study is validated on a modified IEEE 14-bus system, the same approach can be scaled to larger smart-grid and microgrid systems by retraining the models with larger datasets on more buses, distributed generators, and operating conditions.

The proposed HRF–CNN model has a number of strengths when applied to the smart-grid fault diagnosis problem. First, it offers a multi stage diagnostic solution in a single architecture. Second, the framework demonstrates good results in binary (fault and mode detection) and multi-class (bus and fault-type classification) tasks, which indicates its accuracy in addressing the problem of complex classification. Third, it can effectively diagnose contemporary smart grids that are both grid-connected and islanded and can be used at different grid dynamics.

Although these are the strengths of the present study, it has certain limitations because of the dependency of the datasets and validating the study through simulation. Moreover, the framework is mostly based on three-phase current signals and the addition of some other features like voltage or frequency can also be implemented to improve performance. It is also observed that certain fault types, particularly ABC and ABCG, remain more challenging to classify due to similarity in their transient characteristics. Finally, the current approach does not explicitly model conventional protection devices such as impedance relays, and integration with relay-based decision mechanisms remains an important direction for future research.

6. Conclusions

This study presented fault diagnosis framework for modern smart grids by employing hybrid intelligent technique which combine the advantages of ML and DL. The proposed framework utilized the integration of RF Model and CNN for the detection of the initial faults, and to detect the location and nature of the faults. Simulation validation is carried out by different performance criteria including accuracy, precision, recall, f1-score, confusion matrix and analysis of ROC curve. This study confirm that proposed framework using AI are capable of greatly surpassing traditional fault diagnosis strategies in terms of efficiency and reliability of smart grid in both grid-connected and islanded mode. Future research needs to concentrate on implementation in hardware with digital relays or embedded systems, real-time testing, cyber–physical interaction and incorporating optimization. The future work will concentrate on expending the proposed framework such as hardware implementation, real-time relay integration, inclusion of additional features such as voltage and impedance-related quantities, robustness testing under high renewable penetration, and scalability to larger systems.

CRediT authorship contribution statement

Zahid Riaz: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Abdul Qayyum Khan:** Supervision, Project administration, Formal analysis, Data curation. **Salman Habib:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Muhammad Saad:** Writing – review & editing, Validation, Data curation. **Fahad Zaman:** Writing – review & editing, Visualization, Software. **Sultan M. Alghamdi:** Writing – review & editing, Resources, Project administration, Funding acquisition. **Amr Yousef:** Writing – review & editing, Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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